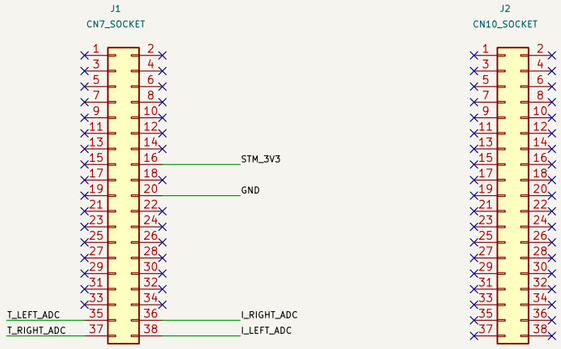
TMR MONITOR SHIELD | RevA.0 DESIGN STARTER

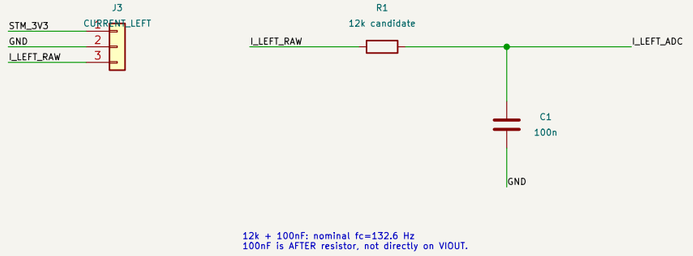
DRAFT – NOT FOR FABRICATION OR CONNECTION
Signal circuits only; current-sensor power path remains off-board.

1. NUCLEO-F446RE MORPHO INTERFACE

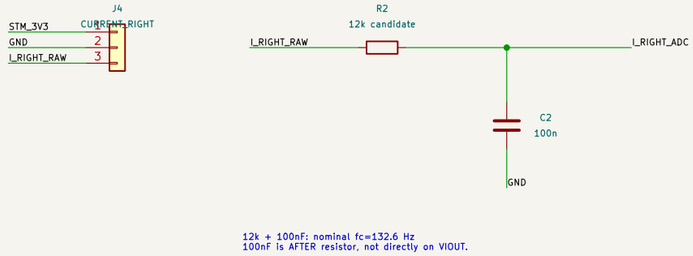


CN7: 16=3V3, 20=GND
35=PC2, 37=PC3
36=PC1/PB9*, 38=PC0/PB8*
* Verify solder bridges; preserve IMU PB8/PB9.
CN10 pins intentionally unused by this shield.
NC marks do not alter the NUCLEO wiring.

2. CURRENT INPUTS – SLOW TREND CANDIDATE

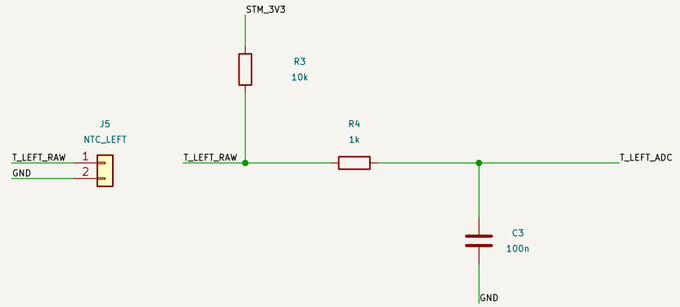


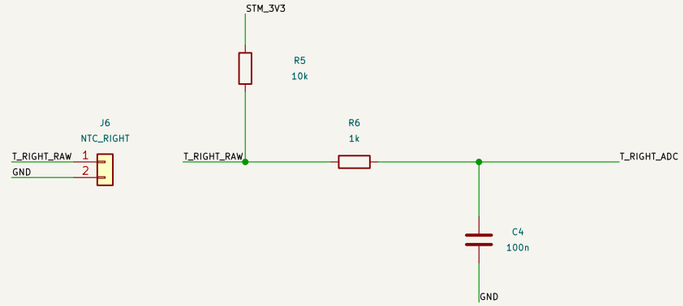
12k + 100nF: nominal fc=132.6 Hz
100nF is AFTER resistor, not directly on VIOUT.



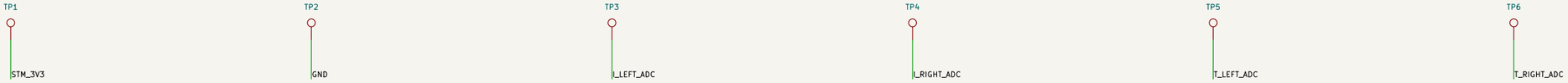
12k + 100nF: nominal fc=132.6 Hz
100nF is AFTER resistor, not directly on VIOUT.

3. NTC INPUTS – 103JT-025 OFF-BOARD





4. TEST POINTS



OPEN ELECTRICAL ITEMS
- Input ESD / fault protection and unpowered backfeed not designed yet.
- 3V3 supply margin, carrier decoupling and cable capacitance to be checked.
- ADC acquisition/leakage budget and sensor disconnection handling remain open.
- No automatic motor stop threshold is defined by this sheet.

OPEN MECHANICAL ITEMS
- Connector models and resistor/capacitor body envelopes are provisional.
- Template header positions only; actual socket height / clearance not verified.
- Keep RESET / USER / jumpers / ST-LINK access and cable strain relief.
- A clean ERC is not proof of these electrical or mechanical conditions.

NOT FOR FABRICATION – protection / mechanical review open

Sheet: /
File: TMR_Monitor_Shield_RevA.kicad_sch

Title: TMR Monitor Shield

Size: A3	Date: 2026-10-23	**Rev: RevA.0-DRAFT**
KiCad E.D.A. 10.0.5		Id: 1/1